

Covid-19 in Saudi Arabia: A Pandemic Data Descriptive Analysis and Visualization

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ABSTRACT

At the end of 2019, humans have been contracted by a new coronavirus, known as Covid-19, that caused a pneumonia outbreak in the Wuhan city in China. As this virus expanded globally outside China, the World Health Organization declared the outbreak as a global pandemic in March 2020. The paper aims to investigate the severity of the outbreak in different regions of Saudi Arabia and the impact of the lockdown through data visualization of different epidemiological metrics of Covid-19, such as the spread growth rate and the case fatality rate. The Data Visualization of the pandemic spread, and fatality was provided with Tableau using the real-time data.

CCS CONCEPTS

• **Human-centered computing** → Visualization; Visualization application domains; Information visualization.

KEYWORDS

Covid-19, Spread growth rate, Case fatality rate, Saudi Arabia

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1 INTRODUCTION

Coronaviruses constitute a large group of viruses that circulate among animals, like pigs. Sometimes, these coronaviruses can

emerge and affect people as well, which is known as spillover. The infection by these viruses can result in serious, even fatal, illnesses affecting respiratory function such as severe acute respiratory syndrome (SARS-CoV) and the Middle East respiratory syndrome (MERS-CoV) [1]. On 31 December 2019, the national authorities in China officially informed the World Health Organization (WHO) about a novel coronavirus (SARS-CoV-2) that caused a pneumonia outbreak in Wuhan City [2]. Although the Chinese authority placed Wuhan city on lockdown, the number of confirmed cases did not cease to increase daily in several regions in China and other countries [3]. On 11 March 2020, WHO declared the outbreak as a global pandemic as the virus has rapidly reached more than 200 countries and claimed thousands of lives [4].

In Saudi Arabia, the Ministry of Health (MoH) has declared the first confirmed Covid-19 case on 2 March 2020 [5]. To minimize the spread of the virus, the Saudi government has applied several strict measures on social movement, social and religious gatherings, travel, and businesses [6]. For instance, a 24-hour curfew was implemented on most of the country, and a lockdown was enforced on heavily affected cities such Riyadh, Mecca, Jeddah, and Medina. Moreover, active and massive testing was launched on 15 April to evaluate the Covid-19 spread rate in major regions. On 21 June 2020, the curfew was fully lifted across the country, and strict health protocols and procedures were issued and required by all activities. However, new confirmed cases have been reported daily as the outbreak is still ongoing. Most of the cases are related to people who failed to follow the precautionary measures including social distancing, wearing face coverings, and practicing hand hygiene [7]. Besides, Saudi Arabia has recorded over 75% of all Gulf countries confirmed Covid-19 cases.

A considerable amount of data has been collected about the confirmed, recovered, and death cases in different regions. Analyzing this data would help assess the outbreak risks and implement appropriate protective measures to contain the spread of the virus and restrain its fatality. Several works were concerned with the analysis of Covid-19 data, aiming for forecasting and prediction (e.g. [3]; [8]–[10]). Lastly, data visualization was utilized in several studies within the context of Covid-19 data (e.g. [11]–[13]). In Saudi Arabia, the Ministry of Health provided an online dashboard [14] to present visual statistics such as the number of confirmed,

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death, and recovered Covid-19 cases. This article aims to provide an analysis and visualization of the Covid-19 spread in different regions of Saudi Arabia using different epidemiological metrics.

The remainder of this article is arranged as follows. Section 2 describes the epidemiological parameters used in the analysis and visualization of the Covid-19 pandemic in Saudi Arabia. Section 3 presents the visualization dashboard of Covid-19 spread using Tableau. Finally, the conclusion is drawn in Section 4.

2 METHODOLOGY: COVID-19 EPIDEMIOLOGICAL METRICS

Various epidemiological metrics can be used to assess the magnitude of the risk posed by this novel coronavirus outbreak and support healthcare communities in initiating critical actions, decisions, control measures, and public restrictions in time. In the following, we present the Covid-19 metrics used in this paper.

2.1 Spread Growth Rate (SGR)

Merely observing the total or the daily number of confirmed cases does not allow us to analyze the growth of the cases number. The weekly spread growth rate gives a slightly more accurate view of whether the pandemic is increasing. The weekly SGR on a specific date measures the ratio change in the number of confirmed cases between two consecutive weeks as follows:

$$SGR_i = nbC_i / nbC_{i-1}$$

where nbC_i is the number of confirmed cases at the i th week from the start of the pandemic.

2.2 Case Fatality Rate (CFR)

The Case Fatality Rate represents the percentage of cases that result in death, and it can be helpful to grasp the severity of a disease. A prior estimate by the WHO situation report issued on 29 June 2020 had put that number at 5% (calculated with the formula: $100 * \text{deaths} / \text{cases}$). However, it is noted that, since the disease is still ongoing and the outcome, either recovery or death, of a large number of confirmed cases is still unknown, this number is an underestimate of the severity of the outbreak. A correct formula of the CFR would consider that current deaths relate to past confirmed cases as follows [15]: $CFR = 100 * \text{deaths at day } x / \text{cases at day } \{x - T\}$ where T is the average time from case confirmation to death. In [16], the authors have provided an estimate of $T=8$ days based on observation of the first Covid-19 cases in Wuhan. In this study, we use $T=7$ days, as follows:

$$CFR_i = 100 * \sum_{t=s}^{t=i} nbD_t / \sum_{t=1}^{t=i-1} nbC_t$$

where CFR_i is the Case Fatality Rate at the i^{th} week ($i = s, \dots, K$). s is the lag where the first death appears. The dividend is the cumulative of the number of deaths nbD_t of the current and preceding weeks ($t = s, \dots, i$). The divisor is the cumulative of the number of cases nbC_t of the preceding weeks ($t = 1, \dots, i - 1$).

2.3 Population Characteristics

It can be insightful to discern the number of confirmed/death cases related to the population in a specific area, which would considerably clarify the disease's spatial exhibition. In this paper, the Case per Population Indicator (CPI) is the number of confirmed cases per 10K people, and the DPI is the number of deaths per 10K population.

$$CPI = nbC \times 10000 / Pop \quad \text{and} \quad DPI = nbD \times 10000 / Pop$$

where nbC (resp. nbD) is the total number of confirmed (resp. deaths), and Pop is the size of the population.

The Case per Area Indicator (CAI) is another valuable indicator to analyze the relationship between the average population density and the disease spread to visualize the areas that are drastically hit by the pandemic. It represents the average density of cases in a specific region ($Area$), as $CAI = nbC / Area$

3 RESULTS: VISUALIZATION DASHBOARD

The first phase of this study consists of data collection and pre-processing. The second phase is the analysis of Covid-19 metrics. The third phase consists of visualizing the Covid-19 spread using Tableau.

3.1 Dataset

The dataset, collected from the Saudi MoH [14], contains three main factors: the number of cases, the number of deaths, and the number of recovered for each city reported daily from the first confirmed case on 2 March until 28 June 2020. From its onset, the number of confirmed cases has steadily increased with the most significant number of cases declared in one day was 4919 on 16 June 2020, reaching a total of 186,436 confirmed cases and 1,599 deaths recorded on 28 June 2020. The epidemic data were aggregated daily by seven days' periods to compute the three factors for the 13 regions of the kingdom, as shown in Table 1.

The most affected regions are Riyadh, Mecca, Medina, and Eastern Province, where combined make up for 75% of the total population. In these regions, the cases total was 167,619, which is 89.9% of the cumulative number of confirmed cases nationwide. Mecca and Riyadh regions lead with over 107,878 confirmed cases (57.9%), making sense since over 51% of 35 million live in these two regions. Moreover, the number of infected cases between the four regions is correlated with their total population. Mecca region is home to over 9.2 million people, while the population of the Medina region, on the other hand, is just over 2.2 million people.

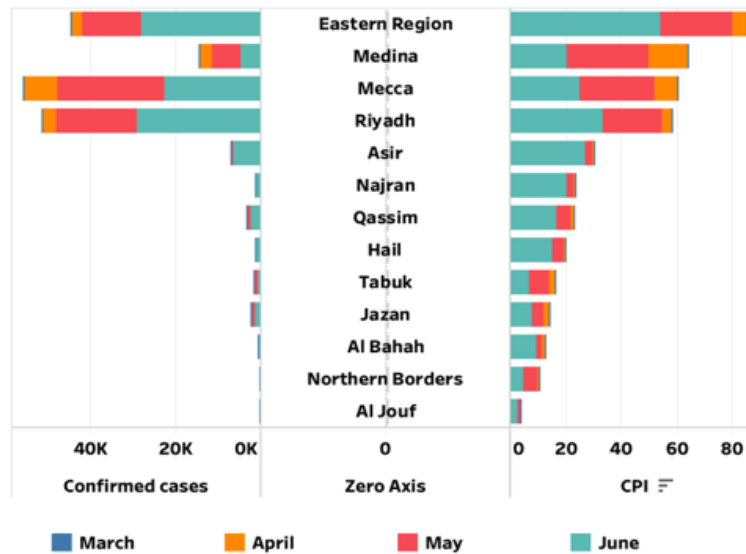
3.2 Covid-19 Metrics Analysis

The epidemic metrics presented in the previous section are analyzed to show how the pandemic affects Saudi Arabia. All the figures included in this paper are extracted from the developed dashboard using Tableau.

3.2.1 Population Characteristics Analysis. Figure 1 shows the CPI for all the 13 regions over March, April, May, and June. The Eastern Province displays a more significant CPI value than the country-wide average: 85 as opposed to 53. However, the Mecca region, the most affected, has a smaller value of the indicator than the Eastern Province (60). In addition, it is shown in Table 1 that the Eastern Province density is half of that of Medina's region, whereas the

Table 1: Total of confirmed and death cases for all the regions on 28 June 2020 and the population density.

Region	Population (Pop)	Area (Km ²)	Density (Pop/km ²)	<i>nbC</i>	<i>nbD</i>	<i>DPI</i>	<i>CPI</i>	<i>CAI</i>
Al Jouf	545,073	100,212	5.44	202	2	0.036	3.71	0.002
Northern Borders	392,850	111,797	3.51	409	10	0.254	10.41	0.0036
Al Bahah	509,872	9,921	51.39	641	6	0.117	12.57	0.065
Jazan	1,678,604	11,671	143.83	2,363	24	0.143	14.08	0.202
Tabuk	973,993	146,072	6.67	1,560	17	0.174	16.02	0.011
Ha'il	749,483	103,887	7.21	1,478	3	0.04	19.72	0.014
Qassim	1,523,723	58,046	26.25	3,521	14	0.092	23.11	0.061
Najran	623,544	149,511	4.17	1,470	1	0.016	23.57	0.01
Asir	2,367,312	76,693	30.87	7,173	15	0.063	30.30	0.09
Riyadh	8,841,619	404,240	21.87	51,849	347	0.392	58.64	0.13
Mecca	9,214,966	153,128	60.18	56,029	921	0.999	60.80	0.37
Medina	2,290,398	151,990	15.07	14,710	90	0.393	64.22	0.10
Eastern Province	5,263,765	672,522	7.83	45,031	147	0.279	85.55	0.07
Total	34,975,201	2,149,690	384.29	186,436	1,599	3.00	53.31	0.087

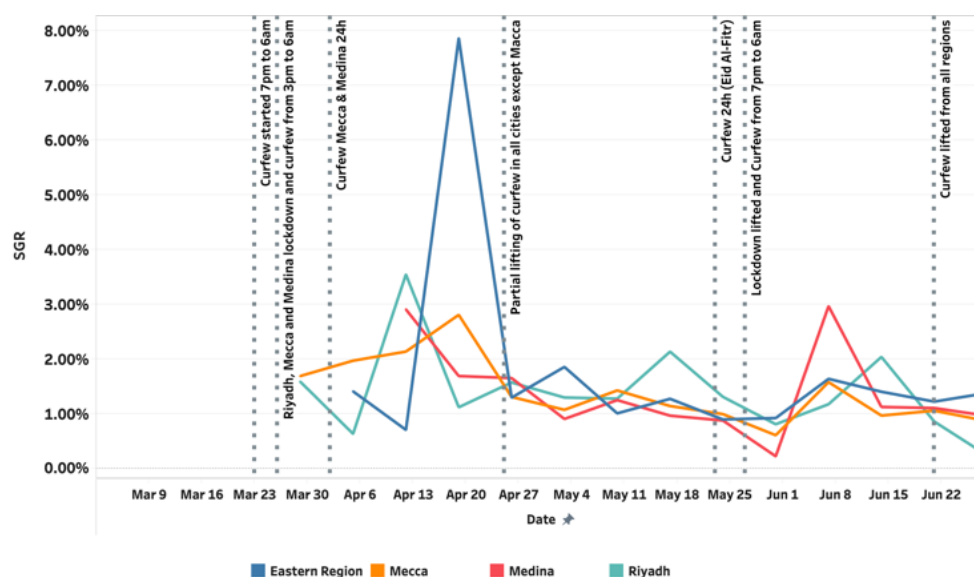
**Figure 1: Covid-19 Cumulative confirmed cases versus the CPI**

total of confirmed cases is three times higher. However, a definitive link between the average population density and the spread of the infection cannot be confirmed. Hence, we choose to use the *CAI*, which may prove to be adequate. This indicator's value is meager in all the regions, with the most substantial computed value found in Mecca's region (0.37 confirmed case/km²). However, we noticed that the total confirmed cases are concentrated in the major cities of Riyadh, Medina, Mecca, Jeddah, and Dammam, as shown in Table 2. Riyadh city alone has more than 88.5% of the cases in Riyadh's region. The *CAI* of Riyadh, 24 confirmed cases per Km², is higher than all other cities, leading to a more prolonged temporal manifestation of the infection.

3.2.2 Growth Spread Rate Analysis. It is very important for the high authority to know how fast the disease is spreading to make decisions. Figure 2 shows the *SGR* for the four major affected regions. As noticed, the first highest *SGR* value (> 3%) was reached in the Eastern Region in the second week of the disease commencement (15 March). This value led the high authority to suspend the international flight and Government workplace attendance after being suspended from all schools and universities. After one week (22 March), the *SGR* exceeded 2% in Mecca's region, which caused the suspension of domestic flights and other means of transport. When the disease attained Riyadh's region with a 2% rate, the authority imposed a curfew (7 pm – 6 am). Later, a lockdown and a national curfew (3 pm – 6 am) were imposed in the Riyadh, Mecca,

Table 2: The CAI of the five major infected cities

Region	Main City	Area (Km ²)	Confirmed Cases	CAI (confirmed cases/Km ²)
Riyadh	Riyadh	1,913	45,910	24
Mecca	Mecca	1,200	24,270	20.22
Mecca	Jeddah	1,600	24,839	15.52
Medina	Medina	589	13,572	23.04
Eastern Province	Dammam	800	11,858	14.82

**Figure 2: The SGR of the four regions throughout the pandemic**

and Medina regions due to the increasing value of the *SGR* (4%). After a prompt decrease of the *SGR* in all regions, the authority partially and gradually released the curfew and lockdown until the complete release on 28 June 2020.

3.2.3 Case Fatality Rate Analysis. The total deaths per 10K population recorded higher values in the four regions than the rest of the country (see Table 1). Mecca's region is the uppermost reaching one death per 10K population. As shown in Figure 3, Mecca's region recorded the highest *CFR* achieving a value above 2%. In contrast, the Eastern Province has recorded the smallest value under 0.5%. On the other hand, Riyadh's region has seen its value increasing from 0.2% to 0.6%. It is also shown that Medina and Mecca achieved the highest *CFR* in April. To reduce the severity of the disease, a 24-hour curfew was imposed on these two regions. By the end of April, the *CFR* decreased below 2% in all four regions, which gradually led to the release of the curfew and the lockdown.

4 CONCLUSION

This study aimed to investigate the spread and severity of Covid-19 in Saudi Arabia regions through data visualization of various epidemic metrics using Tableau. The dataset was collected from 2 March to 28 June 2020. Four main metrics were considered, namely

Spread Growth rate, Case Fatality rate, Case per Area indicator, and Case per Population indicator. The visualization of the Covid-19 spread focused on the four major affected regions such as the Eastern Region, Riyadh, Mecca, and Medina. The *CFR* showed that Mecca and Medina achieved the highest Case Fatality rate in March and April. In contrast, the *SGR* showed that the pandemic was much propagated in Eastern Region in March and April. Moreover, the *CPI* indicated how much the disease is spatially expanded. Furthermore, the *CAI* showed that Riyadh is the most infected city followed by Medina and Mecca cities. The present work can be extended to include the estimation of the reproduction number (*R0*) to show the extent of Covid-19 contamination in Saudi Arabia.

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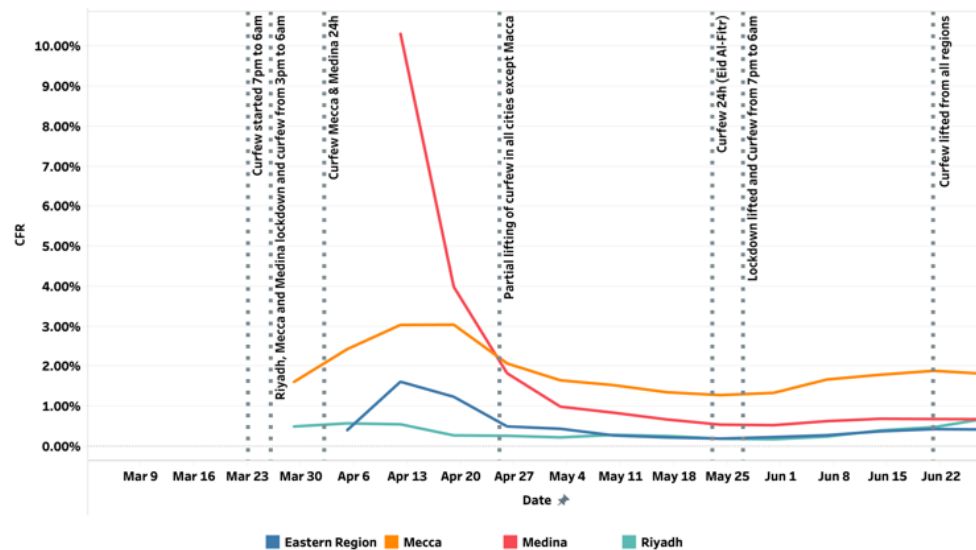


Figure 3: The CFR of the four regions

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